Improving Support Vector Machines for Learning on Imbalanced Data Sets

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Abstract

In this paper, we propose methods to augment SVMs to deal with imbalanced training-data problems posed by many emerging applications (e.g., gene profiling and image understanding). Through a simple example, we show that SVMs can be ineffective in determining the class boundary when the training instances of the target class are heavily outnumbered by the non-target training instances. To tackle this problem, we propose an algorithm which consists of three steps: data cleansing, quasi-bagging, and adaptive conformal transformation. Through empirical studies, we show the improved SVMs to be very effective, improving classification accuracy by about nine percentage points on a $20K$*,* 116*-class image dataset.*

1 Introduction

Support Vector Machines (SVMs) are a core machine learning technology. They have strong theoretical foundations and excellent empirical successes in many pattern recognition applications such as handwriting recognition [8], object recognition [3], and text classification [12]. However, for many emerging applications, such as gene profiling and image understanding, where the training instances of the target class are heavily outnumbered by the rest of the training instances, the class-boundary learned by SVMs can be severely skewed towards the target class. As a result, the false-negative rate can be excessively high in identifying target objects (e.g., a gene disease and a suspicious event) [5, 7, 11, 15, 16, 17, 20], and hence can render the classifier ineffective.

In this paper, we address the problem of imbalanced training classes for SVMs. To understand the causes of the problem and to explore useful methods, we first experimented the problem with two orthogonal approaches: the *algorithmic approach* and the *data-processing approach*. Based on the lessons learned from our extensive empirical studies, we then proposed a hybrid algorithm which uses the algorithmic and data-processing approaches in a complementary way. Our proposed hybrid algorithm consists of three steps.

1. *Data cleansing.* We first use the soft-margin SVM algorithm [18] as our denoising methods to remove unreliable training instances which are located near the separating hyperplane.

2. *Quasi-bagging.* We employ a modified version of *bagging* which includes in each bag a subset of the majority class and the entire minority class.

3. *Adaptive conformal transformation.* We apply an adaptive conformal transformation algorithm to SVMs for training each bag. This transformation gives the neighborhood of minority support vectors a higher spatial resolution, and hence achieves better separation between the classes.

Our experimental results on synthetic and UCI datasets show the hybrid scheme to be very effective in correcting the skewed boundary. On a very challenging 20K, 116-class image dataset, the hybrid scheme improves classification accuracy over SVMs by about nine percentage points (from 53.3% accuracy to 62.5%).

The rest of this paper is organized as follows. Section 2 discusses related work. Section 3 uses

a 2-D example to explain the problem caused by imbalanced training data for SVMs. In Section 4 we describe two basic approaches that we explored, the algorithmic approach and data-processing approach. We then propose a hybrid algorithm to improve SVMs for tackling the imbalanced training-data problem. Section 5 presents the setup and the results of our empirical studies. We offer our concluding remarks in Section 6, as well as suggestions for further studies.

2 Related Work

Researchers have tackled the imbalanced dataset problem mainly in two ways: subsample the majority class, and oversample the minority class. One-sided selection by Kubat [15] is a representative subsampling approach, which removes noisy, borderline, and redundant majority training instances. However, these steps typically can remove only a small fraction of the majority instances, and may not be very helpful in a scenario with a ratio of more than 100 : 1 which is becoming common in many emerging applications. Multi-classifier training proposed by Chan and Stolfo [5], and Bagging proposed by Breiman [4], are two other subsampling methods. These methods do not deal with noisy and borderline data directly, but use a large ensemble of subclassifiers to reduce prediction variance.

Oversampling is the opposite of the downsampling approach [20]. It duplicates or interpolates minority instances in the hope of reducing the imbalance. The over-sampling approach can be considered as a "phantom-transduction" method. It assumes the neighborhood of a minority instance to be negative, and the instances between two minority instances negative. However, assumptions like these can be unreliable. A better approach that has sound theoretical backing is conformal transformation (discussed shortly), which changes the resolution around each training instance.

All sampling methods deal with data. An entirely orthogonal approach is algorithmic, which modifies training algorithms. Nugroho et al. [17] suggest combining competitive learning network and multilayer perceptron as a solution for the class imbalance problem. Kubat et al. [15] create a SHRINK learning system by modifying the decision-tree generator to improve its learning performance on imbalanced datasets. For SVMs, few attempts [13, 16, 19] have dealt with the imbalanced training-data problem. Veropoulos et al. [19] and Lin et al. [16] use different penalty constants for different classes of data. We will explain why this method can be ineffective in Section 4.1. Amari and Wu [1, 21] propose using conformal transformation to change the spatial resolution at the class boundary. We propose a modification to the transformation so that the transformation can adapt to the imbalanced distribution. The experimental results show the adaptive conformation transformation effective.

Our proposed hybrid algorithm to solve the imbalanced training data problem utilizes the strengths of both the algorithmic and data-processing approaches.

3 Boundary Bias and SVMs

In this section, we use a checkerboard example to illustrate the class imbalance problem that SVMs face. We also use the example to explain the causes of the problem.

3.1 Checkerboard Example

A subtle but severe problem that an SVM classifier faces is the skewed class boundary caused by imbalanced training data. To illustrate this problem, Figure 1 depicts a 2D checkerboard example. The checkerboard divides a 200×200 square into four quadrants. The top-left and bottom-right quadrants are occupied by negative (majority) instances, but the top-right and bottom-left quadrants contain only positive (minority) instances. The lines between the classes are the "ideal" boundary that separates the two classes. In the rest of the paper, we will use positive when referring to minority instances, and negative when referring to majority instances.

Figure 2 exhibits the boundary distortion between the two left quadrants in Figure 1 under two different negative/positive training-data ratios, where a black dot with a circle represents a support vector, and its radius represents the weight value α_i of the support vector. The bigger the circle, the larger the α_i. Figure 2(a) shows the SVM class

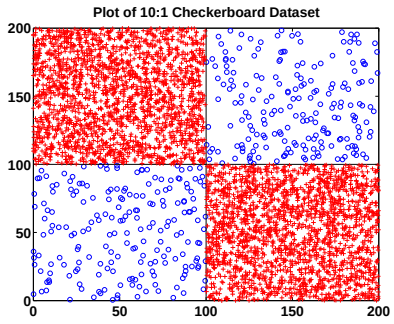
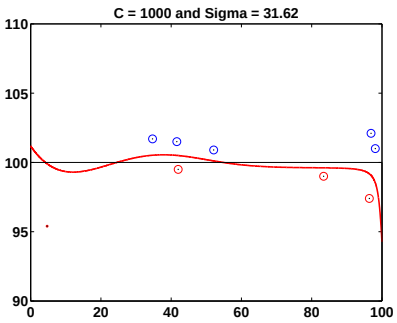
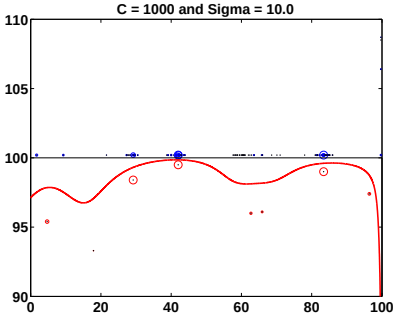


Figure 1. Checkerboard Experiment.

boundary when the ratio of the number of negative instances (in the quadrant above) to the number of positive instances (in the quadrant below) is $10:1$. Figure 2(b) shows the boundary when the ratio increases to $10,000:1$. The boundary in Figure 2(b) is much more skewed towards the positive quadrant than the boundary in Figure 2(a), and hence causes a higher incidence of false negatives.



(a) 10:1



(b) 10,000:1

Figure 2. Boundaries of Different Ratios.

3.2 Causes of Skewed Boundary

To examine the causes, we consider SVMs in the binary classification setting. We are given training data $\{\mathbf{x}_1 \cdots \mathbf{x}_n\}$ that are vectors in some space $\chi \subseteq \Re^d$. We are also given their labels $\{y_1 \cdots y_n\}$ where $y_i \in \{-1, 1\}$. Suppose the training data are linearly separable. The goal of SVMs is to find the optimal hyperplane that separates the training data of the two classes by a maximum margin.

For nonlinearly separable cases, SVMs can project the training data onto a higher dimensional feature space via a Mercer kernel operator K. In addition, Vapnik's soft-margin theory [18] introduces slack variables ξ_i to permit training error. Given $\mathbf{x}_i$, SVMs model class prediction as

$$y_i(\mathbf{w} \cdot \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, ..., n, \quad (1)$$

where $\mathbf{w}$ is the norm to the hyperplane, $|b|/||\mathbf{w}||$ is the perpendicular distance from the hyperplane to the origin, and ϕ is an input-to-feature space mapping function. The optimal solution of SVMs is formulated by maximizing the Lagrangian

$$L_D = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{p} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j), \quad (2)$$

subject to the following constraints:

$$0 \leq \alpha_i \leq C \quad and \quad \sum_{i=1}^{n} \alpha_i y_i = 0, \quad (3)$$

where C is a constant larger than zero used as penalty for soft-margin SVMs. According to the KKT conditions, the value of α_i has three ranges [10]:

$$\alpha_i = 0 \Rightarrow y_i f(x_i) \geq 1 \quad \text{and} \quad \xi_i = 0 \quad (4)$$
$$0 < \alpha_i < C \Rightarrow y_i f(x_i) = 1 \quad \text{and} \quad \xi_i = 0 \quad (5)$$
$$\alpha_i = C \Rightarrow y_i f(x_i) \leq 1 \quad \text{and} \quad \xi_i \geq 0 \quad (6)$$

Once the α_i, $i = 1 \cdots n$ are solved, the class prediction for a testing instance $\mathbf{x}$ can be performed by the following equation:

$$sgn(f(\mathbf{x}) = \sum_{i=1}^{n} y_i \alpha_i K(\mathbf{x}, \mathbf{x}_i) + b). \quad (7)$$

The KKT conditions and Equation 3 reveal a couple of important properties of SVMs. First, only the training instances that have a non-zero α_i (the support vectors) participate in the class decision in

Equation 7. Second, the constraints in Equation 3 show that the total weights α_i of the positive support vectors must be equal to those of the negative support vectors, or

$$\sum_{i=1}^{np} \alpha_i = \sum_{j=1}^{nn} \alpha_j, \tag{8}$$

where np and nn denote the number of positive support vectors and the number of negative support vectors, respectively.

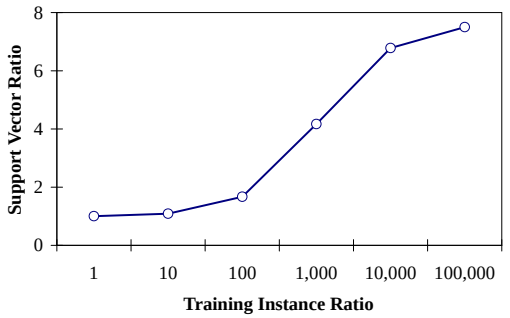


Figure 3. Support Vectors Ratios

To understand the causes of the boundary-skew phenomenon, Figure 3 plots the negative to positive support-vector ratios (nn/np). The x-axis of the figure depicts the ratios of negative to positive training instances, and the y-axis the ratios of negative to positive support vectors. The y-axis shows that the negative to positive support-vector ratio grows as the imbalance between the classes grows. Using the SVM properties and Figure 3, we observe two potential causes for the boundary skew.

1. The imbalanced training-data ratio. At the minority side of the boundary, the positive training data may not always reside as close to the "ideal boundary" as the negative training data do. This can be explained by a simple example. Suppose we randomly draw n numbers based on a uniform distribution between 1 and 100. The larger the value of n, the higher the chance that we draw a number close to 100, though the expected mean of the draws is invariant of the values of n. Thus, the low presence of the positive training instances makes them appear farther from the "ideal boundary" than the negative training instances.

2. The imbalanced support-vector ratio. When inspecting the boundary data in Figure 2, we find the α_i values of the minority class (the positive class) tend to be much larger than those of the majority class, while the number of positive support vectors is substantially smaller. This phenomenon agrees with the constraint presented in Equation 3. As a consequence, the nearest neighborhood of a testing point, especially when it is near the boundary, is likely to be dominated by negative support vectors, and hence the decision function (Eq. 7) is more likely to classify a boundary point negative.

4 Strategies for the Imbalanced Classification

We have studied two approaches for remedying the imbalanced training-data problem: the algorithmic approach and the data-processing approach. The algorithmic approach focuses on modifying the parameters in the class decision function (Section 4.1), and the data-processing approach focuses on processing the training data (Section 4.2). At the end of the section, we present a hybrid algorithm, which takes advantage of the strengths of both approaches.

4.1 Algorithmic Approach

Revisiting the class prediction function of SVMs

$$sgn(f(\mathbf{x}) = \sum_{i=1}^{n} y_i \alpha_i K(\mathbf{x}, \mathbf{x}_i) + b), \tag{9}$$

we can identify four parameters that affect the decision outcome.

- b. The b parameter is the intercept. Changing b shifts the decision boundary.
- α_i. The α_i determines the magnitude of influence of training instance $\mathbf{x}_i$. According to Equation 9, the larger the α_i, the larger the influence of x_i in class prediction.
- K. The kernel function affects the distance computation between a support vector $\mathbf{x}_i$ and a testing instance $\mathbf{x}$. (We assume that K is an RBF function in this paper.)
- γ. The γ parameter determines the width of an RBF function. For instance, when a Gaussian function is employed as the kernel function, the γ parameter decides the width ($\gamma = 1/\sigma^2$) of the

Gaussian function. The smaller the γ, the larger the σ, and hence the larger influence of a support vector in its neighborhood.

4.1.1 Boundary Movement (b)

A naive method is to change b in the SVM decision function. One can set b so that the class-decision boundary can be adjusted as follows:

$$f(x) = sgn(\sum_{i=1}^{n} y_i \alpha_i k(\mathbf{x}, \mathbf{x}_i) + b + \triangle b), \qquad (10)$$

where $\triangle b$ represents how much the boundary is moved. The boundary-movement method is a post-processing method. Intuitively, we can see changing b trades a higher false positive count for a lower false negative one.

4.1.2 Biased Penalties (α_i)

Vapnik [18] introduces soft-margin SVMs by adding the slack-variables ξ_i to relax the hard-margin constraints. The primal formulation of the Lagrangian is as follows:

$$\begin{aligned} L_p &= \frac{\|\mathbf{w}\|^2}{2} + C \sum_{i}^{2} \xi_i^k \\ &\quad - \sum_{i=1}^{n} \alpha_i [y_i (w \cdot \mathbf{x}_i + b) - 1 + \xi_i] - \sum_{i=1}^{p} \mu_i \xi_i, \end{aligned}$$

where $k = 1$ or 2, $\alpha_i \geq 0$, and $\mu_i \geq 0$. The penalty constant C represents the trade-off between the empirical error and the margin.

Veropoulos [19] suggests using different penalty factors C^+ and C^- for positive and negative classes, reflecting their importance during training. Therefore, the L_p formulation has two loss functions for two types of errors.

$$\begin{aligned} L_p &= \frac{\|\mathbf{w}\|^2}{2} + C^+ \sum_{\{i|y_i=+1\}}^{n_+} \xi_i^k + C^- \sum_{\{j|y_j=-1\}}^{n_-} \xi_j^k \\ &\quad - \sum_{i=1}^{n} \alpha_i [y_i (w \cdot \mathbf{x}_i + b) - 1 + \xi_i] - \sum_{i=1}^{p} \mu_i \xi_i. \end{aligned}$$

If the SVM algorithm uses an L_1 norm (k=1) for the losses, its dual formulation gives the same Lagrangian as in (Eq. 2), but with different constraints on α_i as follows:

$$0 \leq \alpha_i \leq C^+, \; if \; y_i = +1, \quad and \qquad (11)$$
$$0 \leq \alpha_i \leq C^-, \; if \; y_i = -1. \qquad (12)$$

It turns out that this biased-penalty method does not help SVMs as much as expected. From the KKT conditions (Eq. 4), one can see that C imposes only an upper bound on α_i, not a lower bound. Increasing C does not necessarily affect α_i. Moreover, the constraint in Equation 3 imposes equal total influence from the positive and negative support vectors. The increases in some α_i at the positive side will inadvertently increase some α_i at the negative side to satisfy the constraint. These constraints can make the increase of C^+ on minority instances ineffective.

4.1.3 γ Scaling

Because SVMs do not assume prior knowledge of the density of the training data, they employ the same γ for all instances. However, for imbalanced training data, the minority class may deserve greater influence in their neighborhood because of their scarcity. One may modify Equation 7 to shape the kernels differently for different classes. Take the Gaussian kernel as an example. We can use different σ for different classes, or we can rewrite the kernel in Equation 7 as

$$K(\mathbf{x}, \mathbf{x}_i) = \exp(-\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{2\sigma_j^2}),$$

where j is the class suffix.

4.1.4 Adaptive Conformal Transformation (K)

Amari and Wu [1] propose modifying the kernel function by changing the Riemannian geometrical structure induced by the function. This method utilizes a conformal mapping to enlarge the spatial resolution around the separating boundary surface in the feature space, such that the SVMs margin could be increased.

Let S denote the input space and F the feature space. From the geometrical point of view, a nonlinear mapping function $\phi(\mathbf{x})$ defines an embedding of S into F as a curved submanifold. When F is a Euclidean or Hilbert space, this embedding induces a Riemannian metric in the input space [1]. Let $\mathbf{z}$ denote the mapped pattern of $\mathbf{x}$ in F. We have $\mathbf{z} = \phi(\mathbf{x}) = (\phi_\alpha(\mathbf{x}))$. Under this mapping function, a small vector $d\mathbf{x}$ is mapped to

$$d\mathbf{z} = \nabla \phi \cdot d\mathbf{x} = \sum_i \frac{\partial}{\partial x_i} \phi(\mathbf{x}) dx_i. \qquad (13)$$

The Riemammian distance ds [1] is defined as

$$ds^2 = |d\mathbf{z}|^2 = \sum_{ij} g_{ij}(\mathbf{x}) dx_i dx_j, \quad (14)$$

where $g_{ij}\mathbf{x})$ is a Riemannian metric induced in S and defined as

$$g_{ij}(\mathbf{x}) = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x'_j} K(\mathbf{x}, \mathbf{x}')|_{\mathbf{x}'=\mathbf{x}}. \quad (15)$$

It represents how a local area is magnified in F under the mapping $\phi(\mathbf{x})$ [1].

Amari and Wu's idea is to enlarge the magnification factor $g_{ij}(\mathbf{x})$ around the boundary but reduce it around other points by modifying the related kernel function K, according to Eq. 15. They suggest implementing this by a conformal transformation of kernel [1, 21]

$$\tilde{K}(\mathbf{x}, \mathbf{x}') = D(\mathbf{x}) D(\mathbf{x}') K(\mathbf{x}, \mathbf{x}'), \quad (16)$$

where $D(x)$ is a properly defined positive function.

Due to Eq. 15, the new Remannian metric is

$$\tilde{g}_{ij}(\mathbf{x}) = D(\mathbf{x})^2 g_{ij}(\mathbf{x}) + D_i(\mathbf{x}) D_j(\mathbf{x}) + 2 D_i(\mathbf{x}) D(\mathbf{x}) K_i(\mathbf{x}, \mathbf{x}), \quad (17)$$

where $K_i(\mathbf{x}, \mathbf{x}) = \partial K(\mathbf{x}, \mathbf{x}')/\partial x_i|_{\mathbf{x}'=\mathbf{x}}$ and $D_i(\mathbf{x}) = \partial D(\mathbf{x})/\partial x_i$. Amari and Wu choose $D(x)$ in a way such that $\tilde{g}_{ij}(\mathbf{x})$ has greater values around the decision boundary. A Gaussian function for $D(\mathbf{x})$ is a good choice, since the last term in Eq. 17 is zero.

In practice, since it is unknown exactly where the optimal boundary is located, Amari and Wu [1] use the support vectors to approximate the boundary, and then increase the metric in the neighborhood of the support vectors. The function $D(\mathbf{x})$ is chosen as

$$D(\mathbf{x}) = \sum_{i \in SV} \exp(-\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{\tau_i^2}), \quad (18)$$

where τ_i is a free parameter and summation runs over all the support vectors. Amari and Wu [21] propose a data-dependent way to choose a dynamic τ_i for each support vector,

$$\tau_i^2 = \frac{1}{M} \sum_{\alpha} \|\mathbf{x}_\alpha - \mathbf{x}_i\|^2, \quad (19)$$

where summation runs over M support vectors $\mathbf{x}_\alpha$ that are nearest to $\mathbf{x}_i$.

This dynamic selection of τ_i just considers the spatial distributions of the support vectors [21], however, does not address the problem of boundary bias in the imbalanced training data scenario. In that case, the decision boundary locates much closer to the minority class than to the majority one. We think it is reasonable to assign a larger metric $g_{ij}(\mathbf{x})$ for the boundary area close to the minority class than for that close to the majority class. Therefore, we propose to assign different scaling factors to the τ_i such that there is a larger value for the τ_i of a minority support vector. The new τ_i is chosen as follows:
if $\mathbf{x}_i$ is a minority support vector,

$$\tau_i^2 = \frac{pr}{M} \sum_{\alpha} \|\mathbf{x}_\alpha - \mathbf{x}_i\|^2, pr > 1. \quad (20)$$

But if $\mathbf{x}_i$ is a majority support vector,

$$\tau_i^2 = \frac{nr}{M} \sum_{\alpha} \|\mathbf{x}_\alpha - \mathbf{x}_i\|^2, nr \approx 1. \quad (21)$$

This method considers both the spatial distribution of support vectors and the boundary bias of the imbalanced classification. Our checkerboard experiment (in Section 5) shows that our method improves the boundary bias better than Amari and Wu's method.

4.2 Data-Processing Approach

Processing training data may help alleviate the imbalance training data problem. We discuss two such methods, denoising and quasi-bagging.

4.2.1 Denoising

It is well known that SVMs can be sensitive to noise. Noisy data near the decision boundary can reduce the margin width and hence reduce the generalization capability of the trained classifier. One solution is to remove unreliable training instances. Kubat and Martwin [15] introduce the method of *Tomek links* to remove noisy instances. However, their method is computationally intensive.

Based on the properties of SVMs, we introduce a two-step denoising algorithm after running the soft-margin SVM algorithm.

1. Remove all false positives from the training dataset.

2. Remove unreliable borderline negative instances from the training dataset. According to the KKT conditions 4, they are the instances with the α_i as C.

4.2.2 Quasi-Bagging

Bagging subsamples training data into a number of *bags*, trains each bag, and aggregates the decisions of the bags to make final class predictions. Our *quasi-bagging* algorithm performs subsampling differently than the bagging scheme does. For each bag, quasi-bagging subsamples only the majority class, but takes the entire minority class. The reason for subsampling just the majority class is to improve class balance in each training bag. At the same time, using a large number of bags reduces the variance caused by the subsampling process. Let $\hat{\theta_n}(\mathbf{x})$ denote a predictor function of a p-dimensional vector from data $\mathbf{z} = \{y_i, \mathbf{x}_i)\}_1^n$. The quasi-bagging algorithm can be described as follows:

1. Randomly draw a bootstrap subset (say a bag) $S_b \subset \{\mathbf{z}_i^*\}_1^m$ from the $\mathbf{z}$. The bag has m samples, including all minority examples.

2. Construct its bootstrap predictor $\hat{\theta}_n^*(\mathbf{x})$

3. Compute the aggregated predictor as

$$\hat{\theta_B}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_n^*(\mathbf{x})$$

There are two parameters in the algorithm to tune: m and B. Parameter m decides the sampling rate and parameter B decides the number of bags. The "best" m and B combination is learned through training.

4.3 A Hybrid Algorithm

Based on the merits of the algorithmic and data-processing strategies, we propose a hybrid algorithm for dealing with imbalanced training data sets. This algorithm employs denoising, quasi-bagging, and adaptive conformal transformation. Let $\mathbf{z}$ be the original training set. The algorithm is summarized as follows:

1. Remove from $\mathbf{z}$ noisy negative instances and unreliable borderline negative instances. We use the denoising strategy discussed in section 4.2. The denoising step yields a new training set $\mathbf{z}'$.

2. Down-sample the new $\mathbf{z}'$ into a group of training subsets $\{\mathbf{z}_i' \| i = 1, 2, .., B\}$ according to a sampling rate $s\%$. B and $s\%$ are empirically obtained by experimenting with different combinations.

3. Apply adaptive conformal transformation on each $\mathbf{z}_i'$.

4. Aggregate B decisions by majority voting.

5 Experimental Results

We present the results of an empirical evaluation of the methods presented in this paper. In particular we show that

- The algorithmic methods that adjust parameters b, α_i, and γ are not as effective as the adaptive conformal-transformation method that adjusts K (kernel function) according to the data distribution.
- Our proposed hybrid scheme significantly outperforms the algorithmic and data-processing approaches when they are employed individually. It also consistently outperforms prior works of Kubat et al. [14, 15], 1-NN, and C4.5 in dealing with the imbalanced training-data problem.

Setup: For each experiment, we first select an RBF kernel (Gaussian or Laplacian) that works the best on the dataset, and we use the best parameter settings (C and γ) obtained through using cross-validation.

Synthetic dataset: We use the checkerboard setup described in Section 3. We report the experimental results on two imbalanced datasets with negative/positive training-data ratios of $10:1$ and $10,000:1$. We generate $2,000$ randomly selected testing instances, $1,000$ at each side of the "ideal boundary", as our testing instances.

Real databases: We use two UCI datasets [2] and a 20K, 116-category image dataset.

- The two UCI datasets we experiment with are, *glass* (g7) and *euthyroid*. These sets were used by Kubat et al. in [14, 15]. Our test setup is identical to Kubat's so that we can compare the effectiveness of our algorithms with theirs.
- The image dataset contains 20K, 116 categories of images collected from the Corel Image CDs[1]. Each image is represented by a vector of 144 dimensions including color, texture, and shape features [6]. To

[1]We exclude categories that are not possible to be automatically classified, such as "industry", "Rome", and "Boston", from our testbed. (E.g., the Boston category contains various subjects, e.g., architectures, landscape, and people, of Boston.)

perform class prediction, we employ the one-per-class (OPC) ensemble [9], which trains 116 classifiers, each of which predicts the class membership for one class. The class prediction on a testing instance is decided by voting among the 116 classifiers.

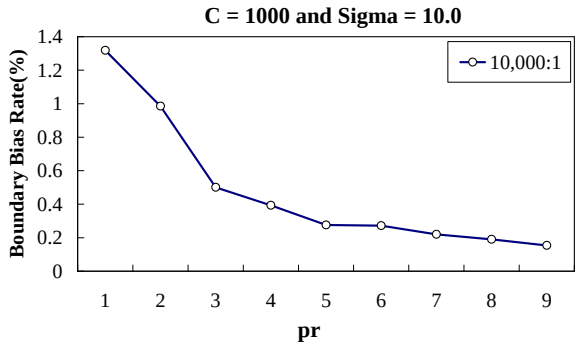


Figure 4. Adaptive Conformal Transformation.

5.1 Evaluating Algorithmic Methods

We compare the prediction accuracy of the four algorithmic methods, boundary movement (BM), biased penalties (BP), γ scaling (GS), and adaptive conformal transformation (ACT), presented in Section 4.1 on the synthetic dataset. Figure 5 shows that on the two selected training data ratios, 10:1 and 10,000:1, BM can achieve only direct tradeoffs between false positives and false negatives. (The GS method produces similar tradeoffs to those of BM, so we do not present it to save space.) Figure 6 presents the results of using the BP method, where we fix $C^- = 1,000$ and vary C^+ between 1,000 and 10,000. The results show that BP does not seriously affect prediction accuracy, which bears out our prediction in Section 4.1.

Next, we test the adaptive conformal transformation (ACT) method. To describe the severity of boundary skew, we use the *boundary bias rate* to measure the effect of skew. The *boundary bias rate* is the percentage of the the area being misclassified over the 200×200 checkerboard area. Figure 4 shows the boundary bias rate (in the y-axis) of the 10,000:1 dataset by fixing $nr = 1$ and varying pr (in the x-axis) from 0 to 4. Note that when $pr = 0$ no conformal transformation is used; $pr = 1$ is Amari and Wu's method. The bias of SVMs without employing conformal transformation at $pr = 0$ is 1.3%. The conformal transformation of Amari and Wu's (at $pr = 1$) reduces the bias to 1%. The ACT scheme can further reduce the bias to as low as 0.2%. This result shows that ACT can adapt to the data and achieve satisfactory correction to the boundary.

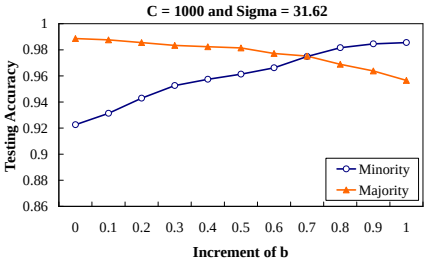
Dataset	C4.5	1-NN	Kubat's	SVMs	Hybrid
g7	92.6	95.2	96.6	92.0	98.1
euthyroid	88.2	60.8	74.0	88.6	92.9

Table 1. UCI-dataset Prediction Accuracy.

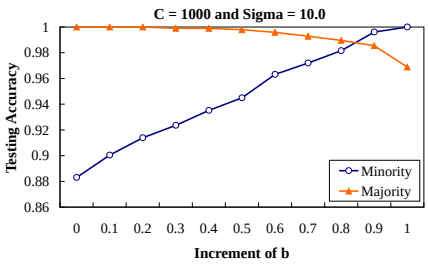
5.2 Evaluating the Hybrid Method

To compare our hybrid scheme with prior methods, we conducted experiments with two UCI datasets and an image dataset. Table 1 reports the experimental results on the two UCI datasets. We use the standard k-fold cross-validation method to obtain class-prediction accuracy for each dataset-algorithm combination. To be consistent with Kubat's performance metric so that we can achieve a fair comparison, we use the Kubat's g-means metric defined as $\sqrt{a^+ \cdot a^-}$, where a^+ and a^- are positive and negative testing accuracy, respectively. The table shows that our hybrid algorithm outperforms Kubat's methods (Kubat et al. used one-side selection for g7 and SHRINK for euthyroid). Our hybrid scheme also outperforms some popular statistical methods such as C4.5, 1-NN, and SVMs by significant margins. The UCI datasets, unfortunately, do not adequately represent the imbalanced training-data challenge that many emerging applications face today. We thus conduct stress tests by using a large, high-dimensional, and diversified image dataset. (The image dataset is discussed in the setup section.)

Table 2 presents the prediction errors of 15 representative categories. We compare five schemes, SVM, biased penalties (BP), adaptive conformal transformation (ACT), quasi-bagging (QB), and the hybrid scheme. First, the unimproved SVMs and BP suffer from the highest average error, 46.7%. We have explained the reason why BP can be ineffective in Section 4.1, and this experiment further validates that this popular method may not be use-

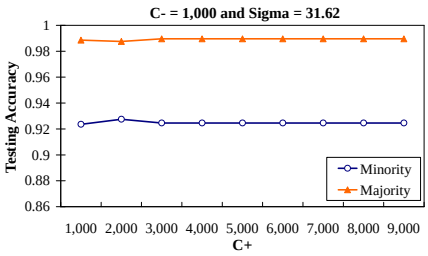


(a) 10:1

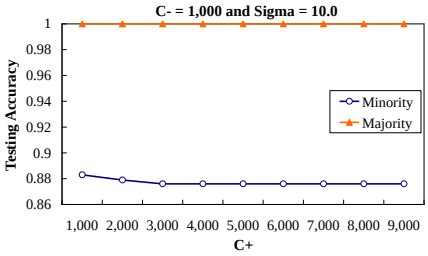


(b) 10,000:1

Figure 5. Testing Accuracy of BM (and GS).



(a) 10:1



(b) 10,000:1

Figure 6. Testing Accuracy of BP.

Category	SVMs	BP	ACT	QB	Hybrid
Cake	80.0	90.0	80.0	80.0	70.0
Camel	100.0	100.0	83.3	83.3	66.7
Cave	47.6	47.6	52.4	57.1	47.6
Cheetah	50.0	50.0	33.3	33.3	33.3
Cityscape	34.7	37.3	41.3	42.7	38.7
Elephant	57.7	53.8	42.3	46.2	38.5
Lizard	69.4	80.6	47.2	50.0	44.4
Mountain	41.9	46.7	55.2	58.1	53.3
Rose	38.9	38.9	27.8	27.8	27.8
Shark	66.7	66.7	33.3	33.3	33.3
Sheep	71.4	71.4	71.4	71.4	57.1
Temple	72.3	53.2	21.3	21.3	21.3
Texture	28.0	32.0	20.0	21.3	18.7
Waterfall	41.0	41.0	34.6	35.9	32.1
Wave	52.0	44.0	40.0	40.0	36.0
Average Error	46.7	47.2	40.1	41.8	37.5

Table 2. Image-dataset Prediction Errors.

ful when employed by SVMs. Next, both ACT and QB[2] schemes improve SVMs by reducing the average error by about five to six percentage points. This result confirms that both ACT and QB are helpful in reducing boundary skew. Finally, the hybrid scheme, which combines data cleansing, QB, and ACT, reduces the average error by 9.2 percentage points (from 46.7% to 37.5%) in prediction accuracy.

6 Conclusion and Future Work

We have proposed a hybrid algorithm, which consists of data cleansing, quasi-bagging, and adaptive conformal transformation, for tackling the imbalanced training-data challenge. Through example,

[2]We select the best sampling rate and the optimal number of bags for the QB scheme through cross-validation. The best sampling rate is 10% and the optimal number of bags is 100. Due to space limitation, we do not present intermediate results here.

theoretical justifications, and empirical studies, we show our proposed scheme to be very effective. On both synthetic and real-world datasets, it improves classification accuracy significantly.

Because many emerging applications, especially the ones in the imagery and biotech areas, face increasingly high-dimensional and imbalanced training data sets, we believe this work is important. Our future work can be expanded in two directions. On the data-processing front, we believe that a more intelligent sampling method could further improve classification accuracy. On the algorithmic front, we plan to study robust algorithms that can work effectively in very high-dimensional spaces, where the dimensionality curse can further exacerbate the problem.

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